

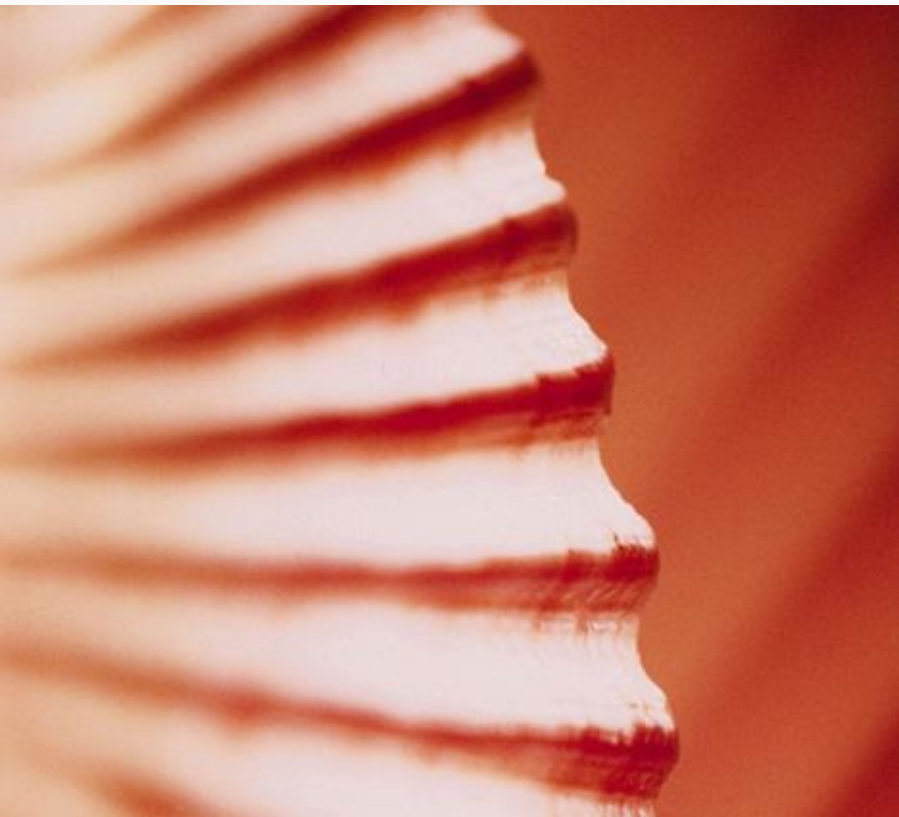
The Meaning of *Understanding* in AI-Laden Science

Siyu Yao (yaosy@sjtu.edu.cn)
Shanghai Jiao Tong University

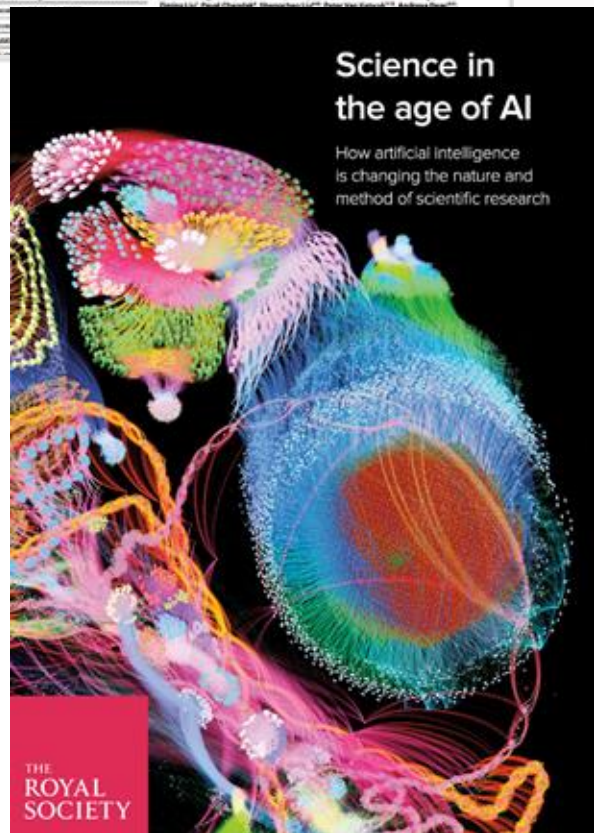
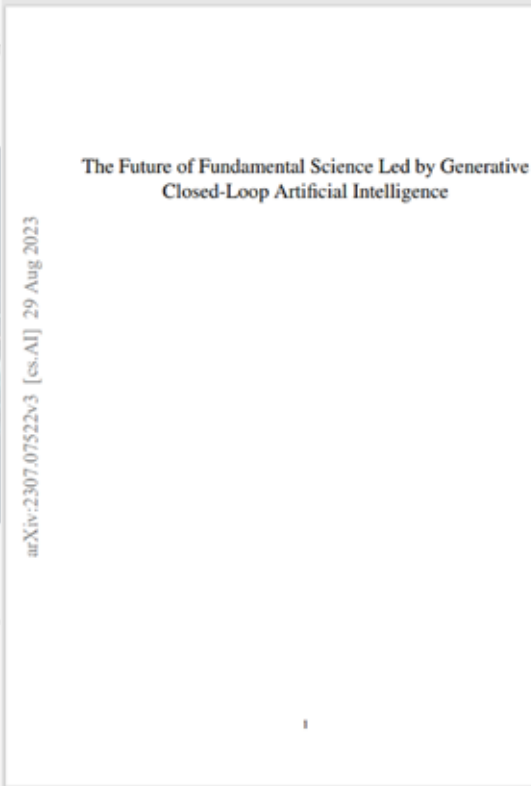
André Curtis-Trudel
(curtisa4@ucmail.uc.edu)
University of Cincinnati



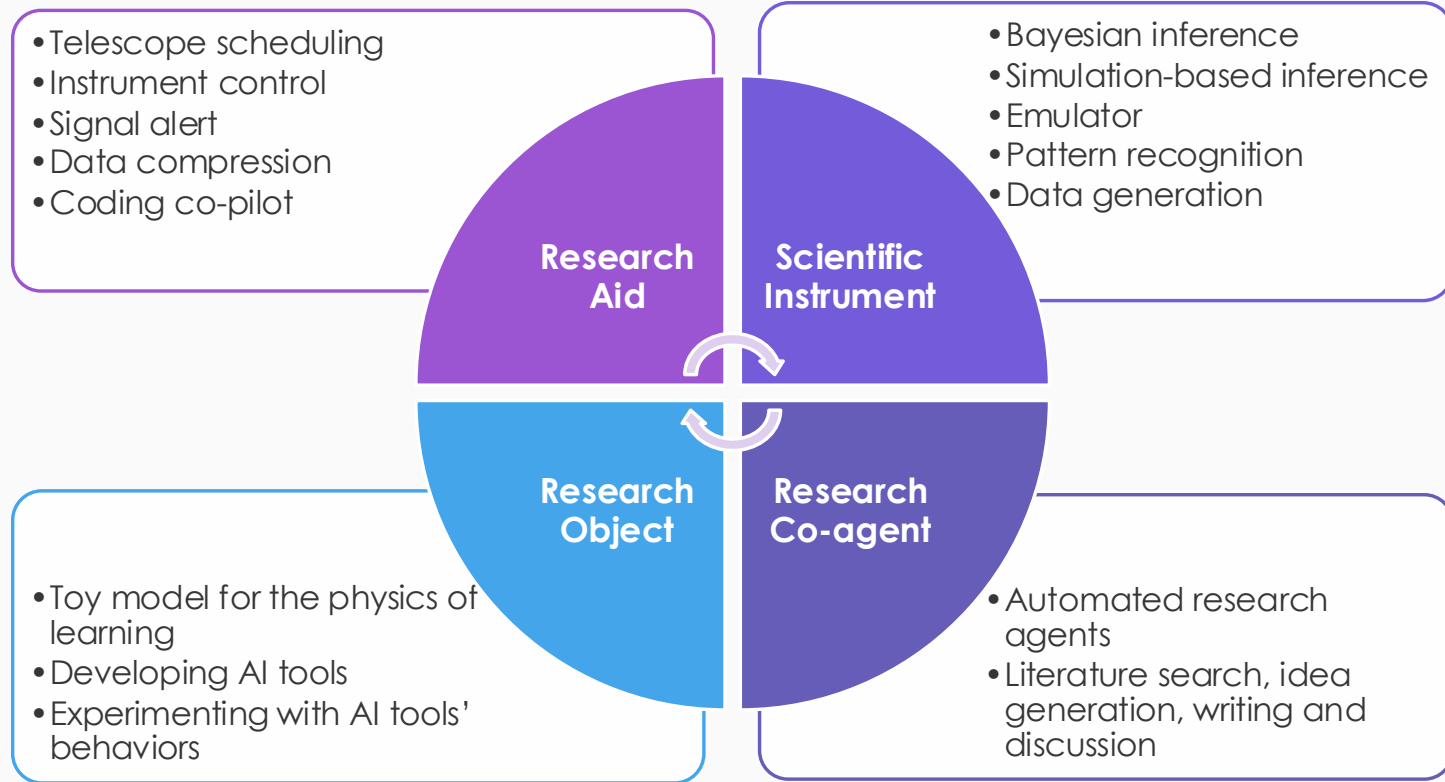
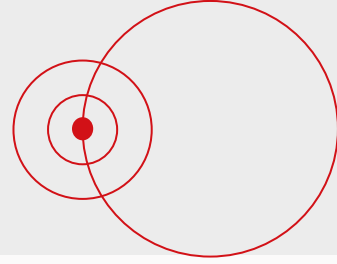
Outline



- 01 AI-Laden Science and the Relevance of Philosophy
- 02 AI, Explanation and Understanding
- 04 Empirical Study of Understanding
- 05 What's Worthy of Pursuit Now?
- 06 Conclusion & Discussion



AI-Laden Astronomy



How is AI changing science?

Editorial | Published: 15 January 2026

AI challenges science to reinvent itself

[Nature Reviews Bioengineering](#) 4, 1 (2026) | [Cite this article](#)

2957 Accesses | 2 Altmetric | [Metrics](#)

As artificial intelligence enters the scientific arena, it not only compels us to rethink the scientific method but also opens the door to reimagine long-standing practices.

Comment | Published: 11 March 2026

What understanding means in AI-laden astronomy

[Yuan-Sen Ting](#) , [André Curtis-Trudel](#) & [Siyu Yao](#)

[Nature Astronomy](#) 10, 468–471 (2026) | [Cite this article](#)

309 Accesses | [Metrics](#)

The British philosopher Mary Midgley once remarked that philosophy is like plumbing: you don't notice it until things start to smell funny. And astronomers are beginning to sniff the air. As AI transforms astronomical research, scientists are confronting foundational philosophical questions about the nature of discovery and understanding – questions that call for collaboration with philosophers of science.

Comment | Published: 17 November 2025

Rethinking the scientific method in the age of AI

[Bernd Roloffs](#), [Ryuji Kato](#), [Melanie L. Hart](#), [Corinna Raimondo](#) & [Mary G. Walsh](#) 

[Nature Reviews Bioengineering](#) 4, 2–3 (2026) | [Cite this article](#)

546 Accesses | 1 Altmetric | [Metrics](#)

 An [Author Correction](#) to this article was published on 25 November 2025

 This article has been [updated](#)

Artificial intelligence might reshape the approach to science, from generating concepts and hypotheses to designing experiments, analysing data and publishing findings. However, it might also introduce risks around data integrity, reproducibility, accountability and bias. Here we provide practical guidance on how to responsibly integrate artificial intelligence into the research pipeline.

Comment | Published: 17 November 2025

Rethinking the scientific article in the age of AI

[Miikka J. Lehtonen](#)  & [Dennis Jancsary](#) 

[Nature Reviews Bioengineering](#) 4, 4–5 (2026) | [Cite this article](#)

385 Accesses | 1 Altmetric | [Metrics](#)

If large language models can replicate academic writing, perhaps it was never as creative as once believed. We propose rethinking the scientific article as a multimodal endeavour. Exploring how words, visuals and other modes can work together helps us imagine new ways of creating and communicating scientific knowledge.

Key *philosophical* questions

Scientific Methodology

How is AI changing scientific methodology? Is AI a tool or a collaborator? What are some promises or biases?

Scientists' Identity

What does it mean to be a good scientist? Is there anything unique to scientists? What is AI literacy?

Scientific Virtues

What are science's aims? Is AI highlighting some while suppressing others? What is novelty?

Ethics

How to build norms of using and crediting AI?
How much work should we outsource to AI?

Community

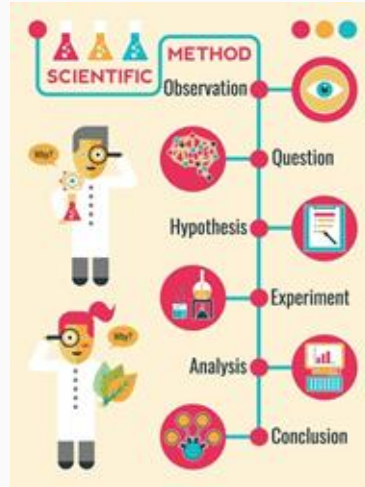
How will AI change community dynamics and pursuit? How will different fields interact?

Things *philosophers* of science do

*Conceptual
engineering*



*Critical examination
of assumptions*



*Abstraction and
generalization*





Lisa Messeri



Molly Crockett

Perspective | Published: 06 March 2024

Artificial intelligence and illusions of understanding in scientific research

[Lisa Messeri](#) ✉ & [M. J. Crockett](#) ✉

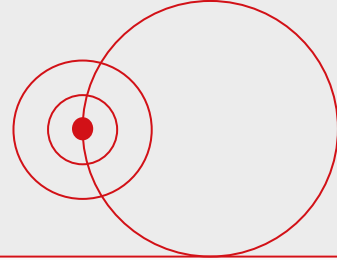
[Nature](#) **627**, 49–58 (2024) | [Cite this article](#)

87k Accesses | **505** Citations | **948** Altmetric | [Metrics](#)

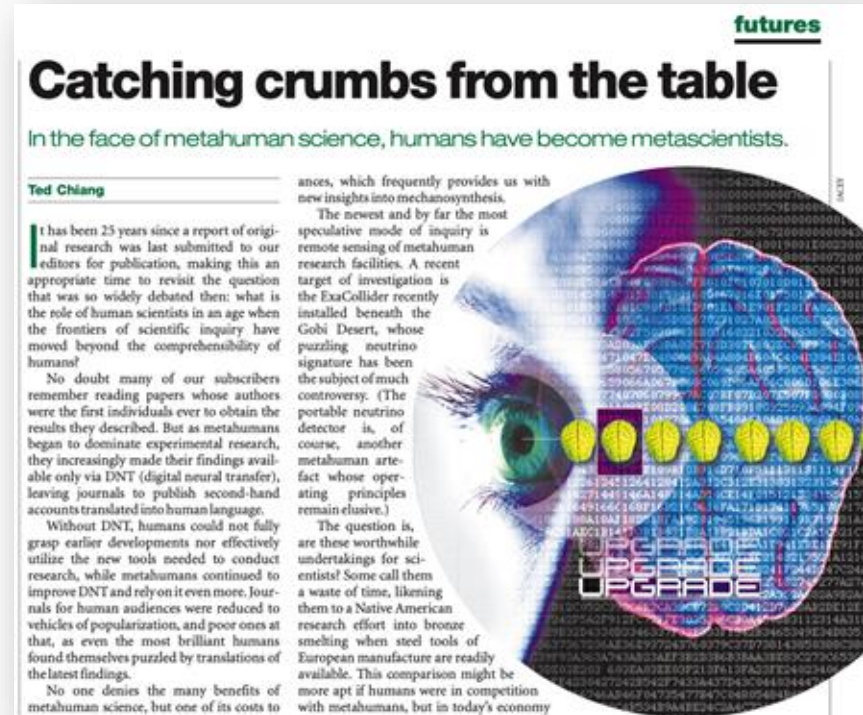
Scientists may mistake AI's outcomes as genuine understanding.

- Illusion of explanatory depth: prediction $\neq$ explanation
- Illusion of exploratory breadth: AI hypotheses $\neq$ all hypotheses
- Illusion of objectivity: AI viewpoints $\neq$ all viewpoints

Challenges to understanding



- **Inscrutable** non-linearity and parameters.
- Failing to answer an agent's **epistemic interest**. (Zednik 2019)
- **Unknown representational assignment** of the target system. (Kawamleh 2021)
- **Link uncertainty**: the lack of evidence supporting the link that connects significant factors in a model to the real causal factors in the target phenomenon. (Sullivan 2022)
- "DNN finds significant features, but the translation of these into scientific concepts is up to scientists' ability and knowledge." (Boge 2022)



Interesting Scientific Idea Generation using Knowledge Graphs and LLMs:
Evaluations with 100 Research Group Leaders

Xuemei Gu^{1,*} and Mario Krenn^{1,*}

¹Max Planck Institute for the Science of Light, Staudtstrasse 2, 91058 Erlangen, Germany

COGNITIVE SCIENCE

Using large-scale experiments and machine learning
to discover theories of human decision-making

Joshua C. Peterson^{1,*}, David D. Bourgin^{1,†}, Mayank Agrawal^{2,3},
Daniel Reichman⁴, Thomas L. Griffiths^{1,2}

Forecasting high-impact research topics
via machine learning on evolving knowledge graphs

Xuemei Gu^{1,*} and Mario Krenn^{1,†}

¹Max Planck Institute for the Science of Light, Staudtstrasse 2, 91058 Erlangen, Germany

Rediscovering orbital mechanics with machine learning

Pablo Lemos^{*1,2}, Niall Jeffrey^{†3,2}, Miles Cranmer⁴, Shirley Ho^{4,5,6,7}, and Peter Battaglia⁸

Understanding from Machine Learning Models

Emily Sullivan 

On scientific understanding with artificial intelligence

Mario Krenn, Robert Pollice, Si Yue Guo, Matteo Aldeghi, Alba Cervera-Lierta,
Pascal Friederich, Gabriel dos Passos Gomes, Florian Häse, Adrian Jinich,
AkshatKumar Nigam, Zhenpeng Yao  and Alán Aspuru-Guzik 

ARTICLE

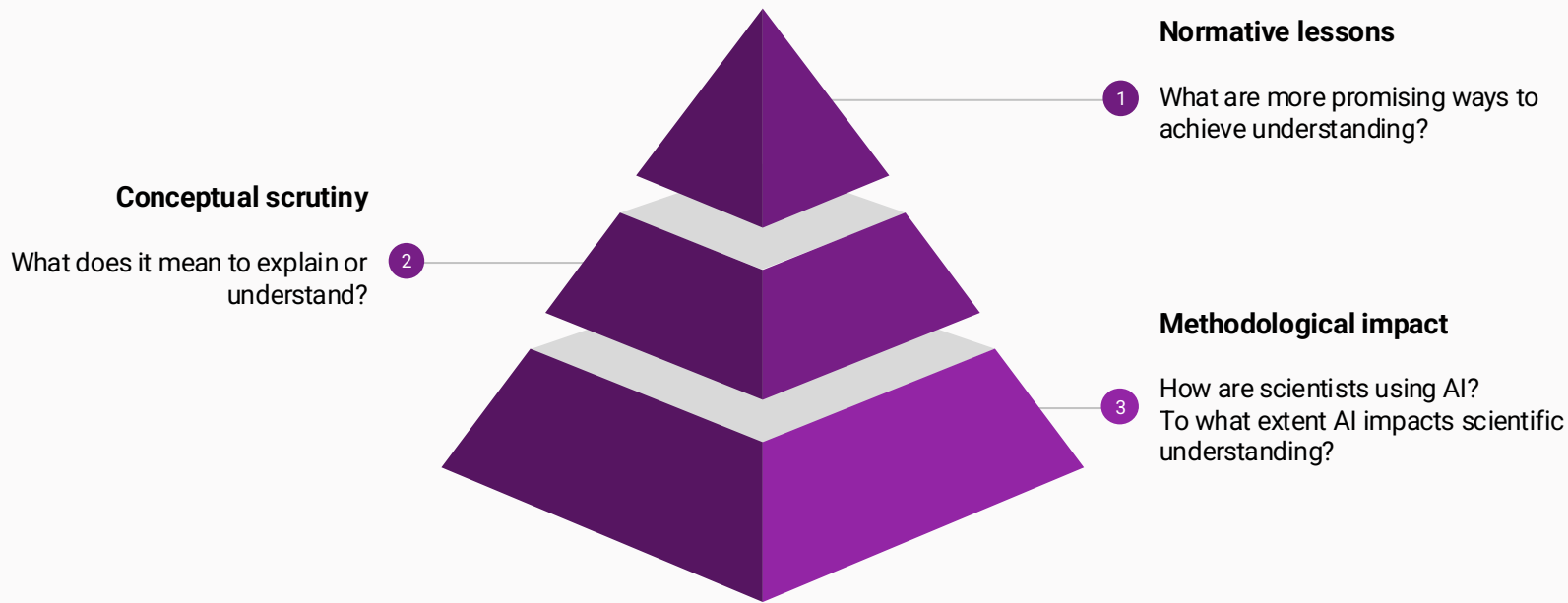
Reliability in Machine Learning

Thomas Grote¹  | Konstantin Genin¹ | Emily Sullivan²

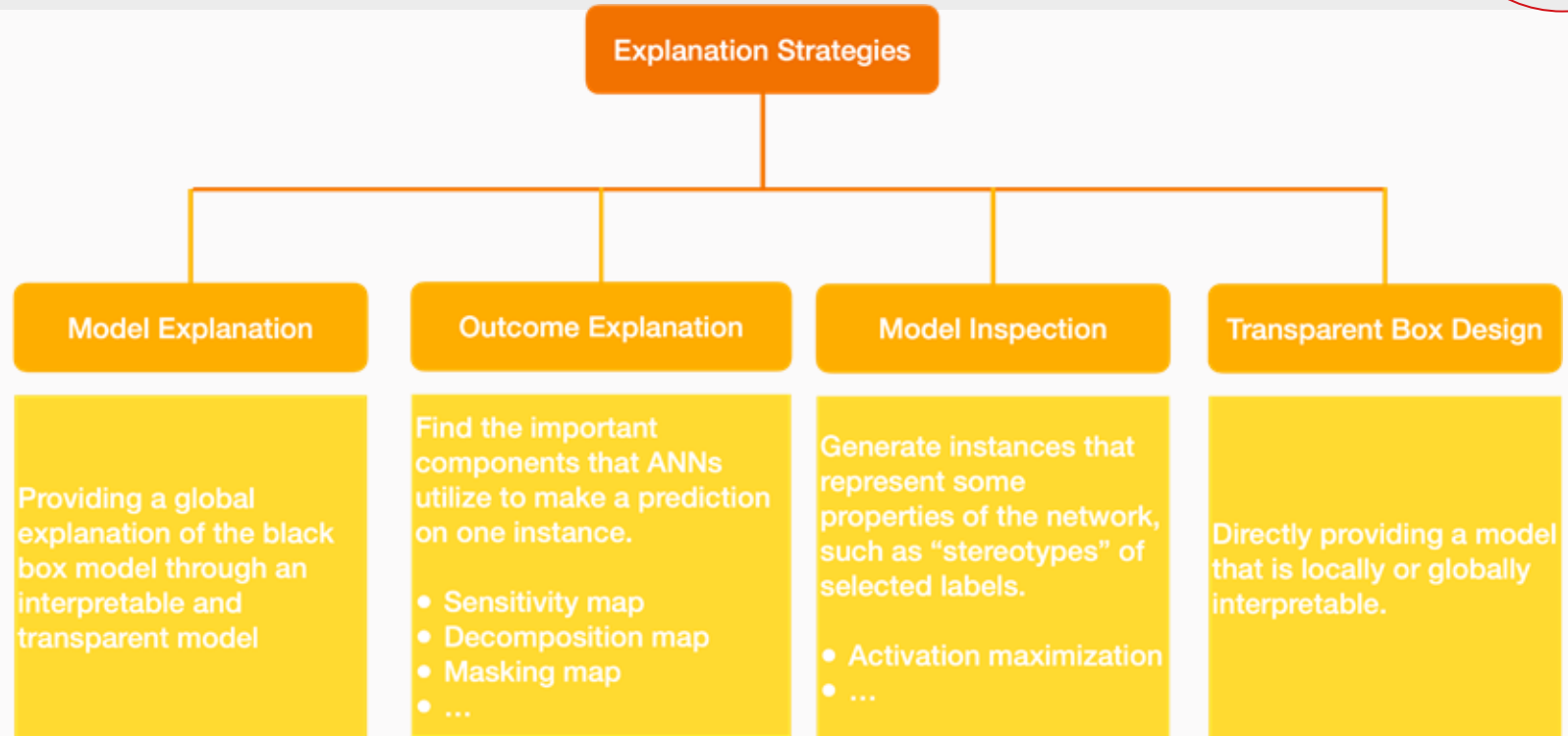
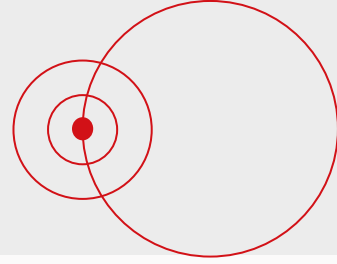
Challenges to understanding

Central concerns:

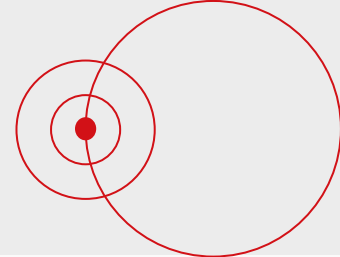
- Can we understand AI's workings and outputs?
- Will AI make us sacrifice scientific understanding?
- What does understanding mean, actually?



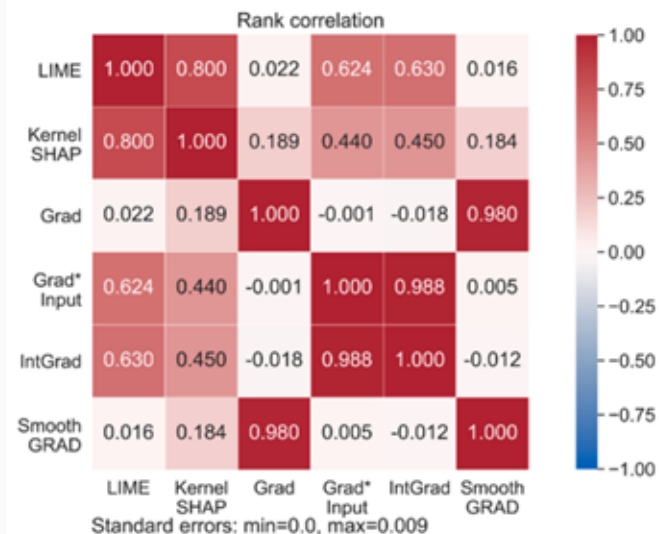
Explanation of AI



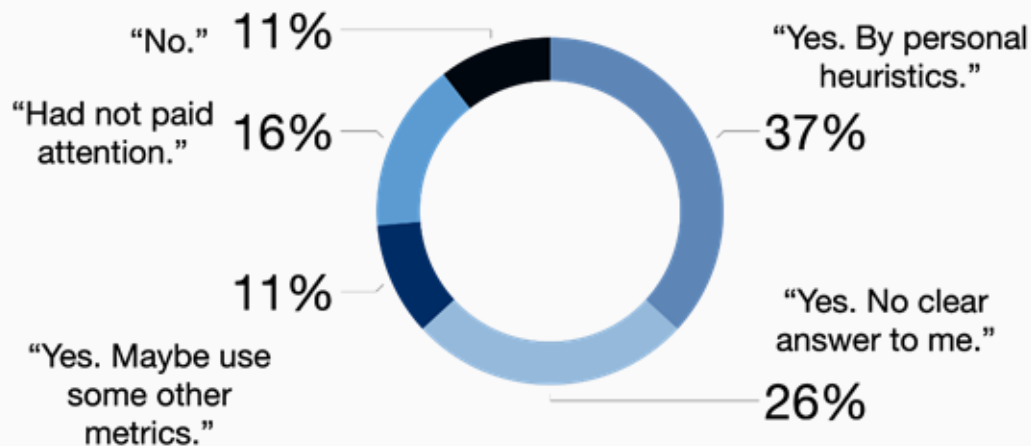
Explanation of AI



Agreement between different XAI methods for image data.



Have you noticed disagreements between explanation outputs? If so, how do you resolve it?



Explanation of AI

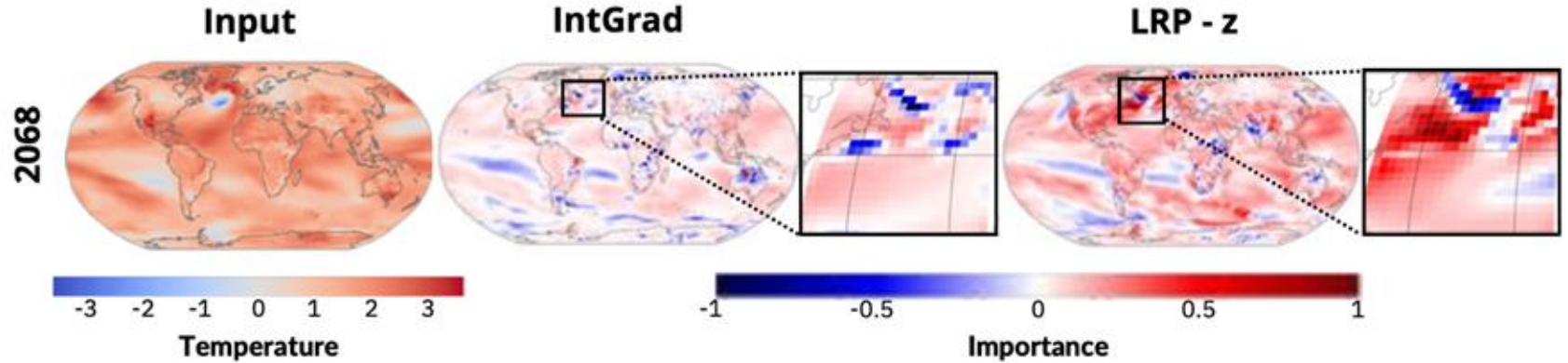
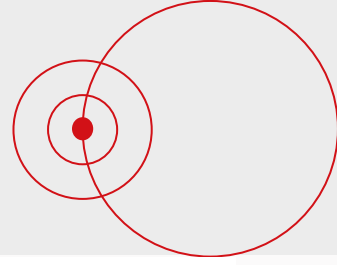
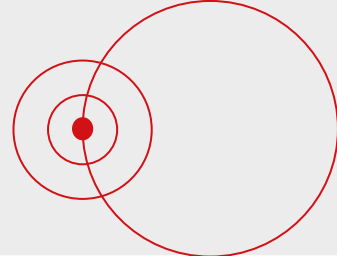


Figure 7: Motivating example visualizing the difference between different XAI methods. Shown are the T2m-temperature map for the year 2068 with the corresponding IntGrad and LRP-z explanation maps of the MLP. For both XAI methods, red means that pixels contribute positively and blue means that corresponding pixels contribute negatively to the predicted class. Next to the explanation maps, a zoomed-in map of the North Atlantic region (NA, 10 – 80°W, 20 – 60°N) is shown, indicating partly different evidences of Integrated Gradients and LRP-z.

Explanation of AI



Explainable Artificial Intelligence (XAI) 2.0: A manifesto of open challenges and interdisciplinary research directions

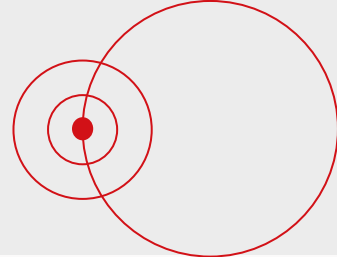
Luca Longo^{1,2},  , Mario Brcic³, Federico Cabitza^{4,5}, Jaesik Choi^{6,7}, Roberto Confalonieri⁸,
Javier Del Ser^{9,10,11}, Riccardo Guidotti¹², Yoichi Hayashi¹³, Francisco Herrera¹¹, Andreas
Holzinger¹⁴, Richard Jiang¹⁵, Hassan Khosravi¹⁶, Freddy Lecue¹⁷, Gianclaudio Malgieri¹⁸,
Andrés Páez^{19,20}, Wojciech Samek^{21,22,23}, Johannes Schneider²⁴, Timo Speith^{25,26}, Simone
Stumpf²⁷

Moving forward?

Abstract

Understanding black box models has become paramount as systems based on opaque Artificial Intelligence (AI) continue to flourish in diverse real-world applications. In response, Explainable AI (XAI) has emerged as a field of research with practical and ethical benefits across various domains. This paper highlights the advancements in XAI and its application in real-world scenarios and addresses the ongoing challenges within XAI, emphasizing the need for broader perspectives and collaborative efforts. We bring together experts from diverse fields to identify open problems, striving to synchronize research agendas and accelerate XAI in practical applications. By fostering collaborative discussion and interdisciplinary cooperation, we aim to propel XAI forward, contributing to its continued success. We aim to develop a comprehensive proposal for advancing XAI. To achieve this goal, we present a manifesto of 28 open problems categorized into nine categories. These challenges encapsulate the complexities and nuances of XAI and offer a road map for future research. For each problem, we provide promising research directions in the hope of harnessing the collective intelligence of interested stakeholders.

Explanation of AI



Or going back?

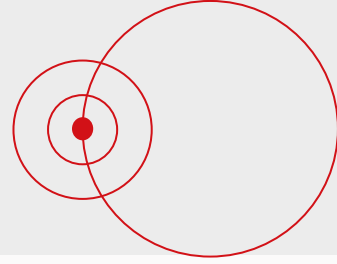
Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead

Cynthia Rudin
Duke University
cynthia@cs.duke.edu

Abstract

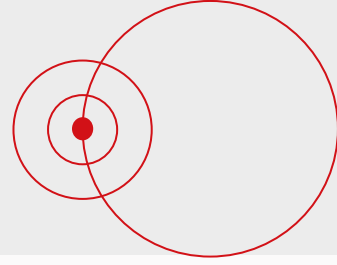
Black box machine learning models are currently being used for high stakes decision-making throughout society, causing problems throughout healthcare, criminal justice, and in other domains. People have hoped that creating methods for explaining these black box models will alleviate some of these problems, but trying to *explain* black box models, rather than creating models that are *interpretable* in the first place, is likely to perpetuate bad practices and can potentially cause catastrophic harm to society. There is a way forward – it is to design models that are inherently interpretable. This manuscript clarifies the chasm between explaining black boxes and using inherently interpretable models, outlines several key reasons why explainable black boxes should be avoided in high-stakes decisions, identifies challenges to interpretable machine learning, and provides several example applications where interpretable models could potentially replace black box models in criminal justice, healthcare, and computer vision.

Explanation and understanding in science



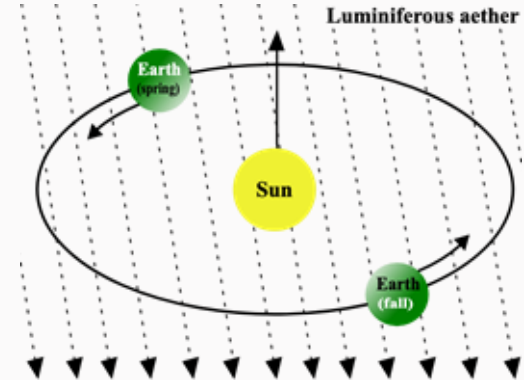
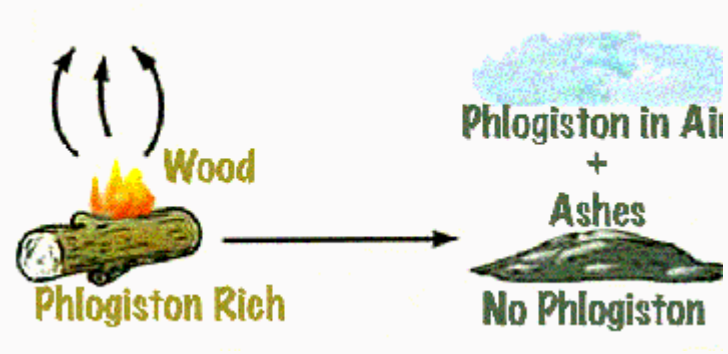
- Does explanation and understanding carry different meanings?
- Can we understand a system without explaining its inner workings in detail?
- When we can explain how a system works, does it mean that we fully understand it?
- Which is the more suitable concept to use if one talks about something's use, significance, implications, etc.?
- What are explanation and understanding good for, respectively?

Explanation and understanding in science

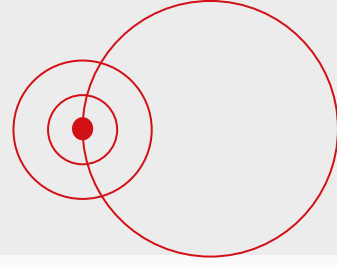


Explanation-first approach: A false explanation may generate a “feeling” of understanding in some people.

Factivity and a proper logical form are important elements of explanation.

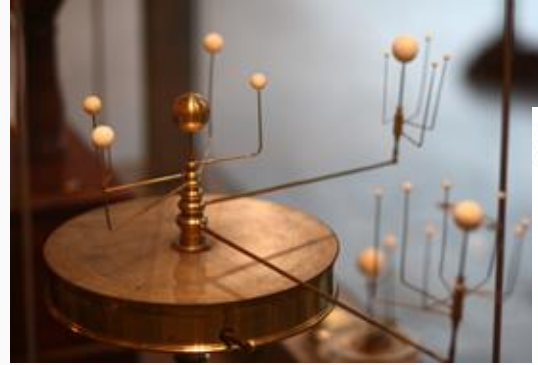


Explanation and understanding in science

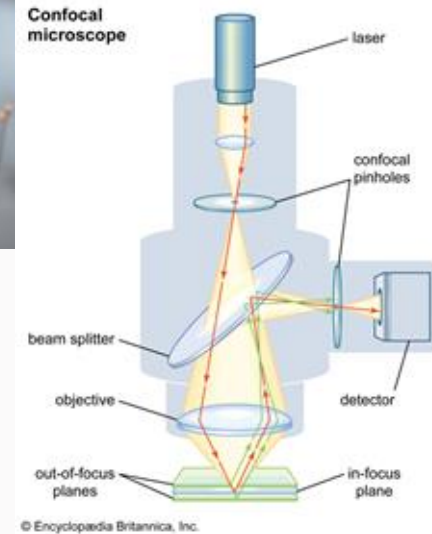
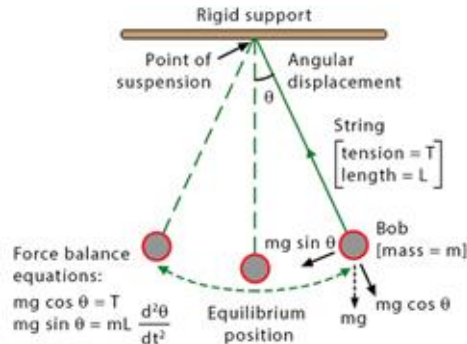


Understanding-first approach:

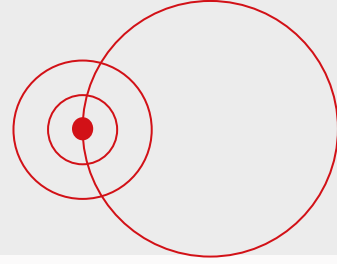
understanding is richer concept;
sometimes more useful and
fundamental; factive explanation is
not necessary for effective
understanding.



Simple Pendulum



Explanation and understanding in science



Explanation

- Of a phenomenon
- Formulaic standards
- Stresses factivity
- External to individuals
- Understanding = knowing a correct explanation?

Understanding

- Of a phenomenon, domain, meanings, how to use something...
- First-personal experience
- Internal standards
- Understanding = a richer concept; explanation is only one vehicle toward understanding?

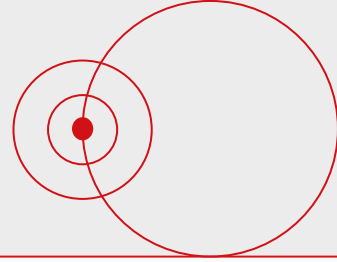
Open questions

Can we understand AI's uses in science without fully explaining its inner details?

Or a good combination of both?

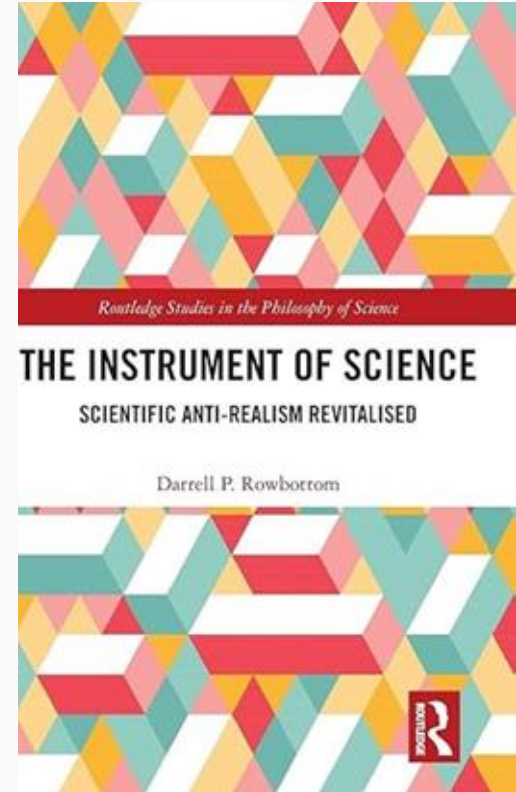
What form of understanding is relevant and helpful?

Different forms of understanding

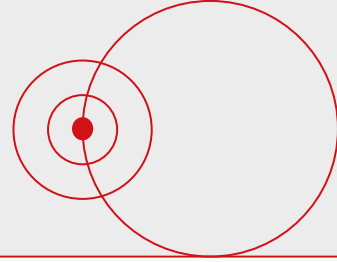


Empirical understanding

- Following Mach and Poincaré: our representation of appearances are empirically adequate and useful.
- “either as artifices for calculation, or to assist our understanding by concrete images, to fix the ideas, as we say.” (Poincaré 1905)
- Grasping: (1) identify how the various elements about a device depend upon and relate to one another; (2) seeing how the device is suitable for one’s ends.

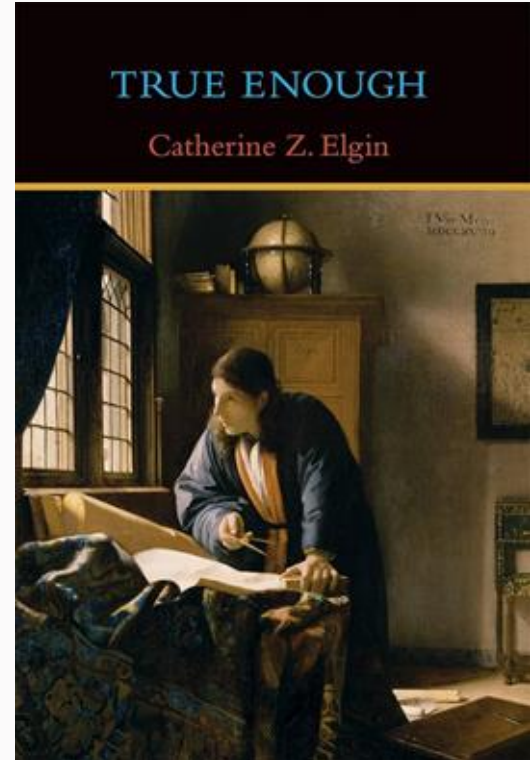


Different forms of understanding

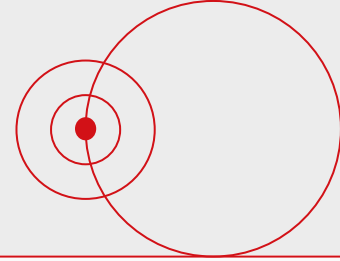


Objectual understanding

- Understanding is not mirroring.
- “To understand **a topic** involves knowing how to wield one’s commitments to further one’s epistemic ends. It involves being able to draw inferences, raise questions, frame potentially fruitful inquiries, and so forth.”
- Understanding may involve “felicitous falsehood”: “an inaccurate representation whose inaccuracy does not undermine its epistemic function.”
- Acceptance of falsehoods: being willing and able to use them for inference or action.



Different forms of understanding

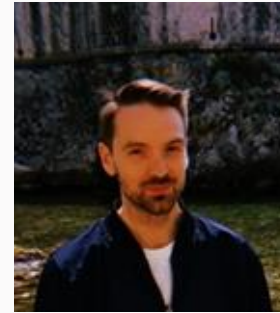


Pragmatic understanding

Philosophical Studies
<https://doi.org/10.1007/s11098-025-02336-6>

A New Account of Pragmatic Understanding, Applied to the Case of AI-Assisted Science

Michael T. Stuart¹ 

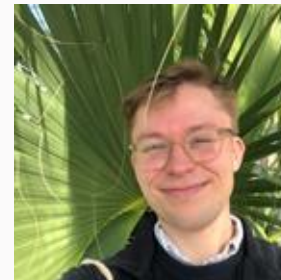


Mike Stuart
(York University)



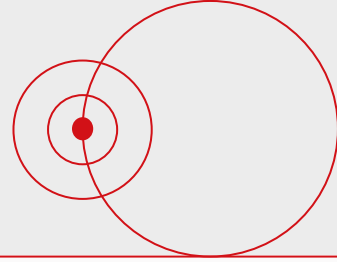
Phillip Kieval
(University of Florida)

Deep Learning as Method-Learning: Pragmatic Understanding, Epistemic Strategies and Design-Rules Phillip Hintikka Kieval and Oscar Westerblad



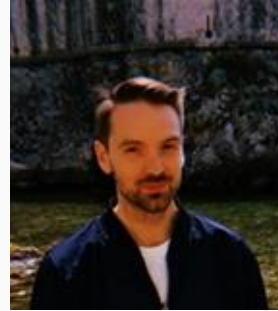
Oscar Westerblad
(University of Iceland)

Different forms of understanding



Pragmatic understanding

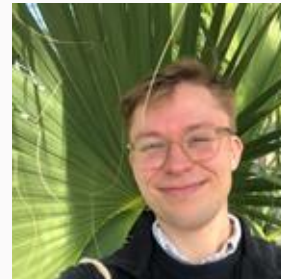
- The understanding of a **scientific device or method**, as opposed to a phenomenon to be studied by this device.
- **Know-how** based: Knowledge about how to use the device to achieve one's aims effectively and successfully, often also including the ability to do so.
- Can include explicit knowledge, tacit skills, strategies, and expertise...



Mike Stuart
(York University)

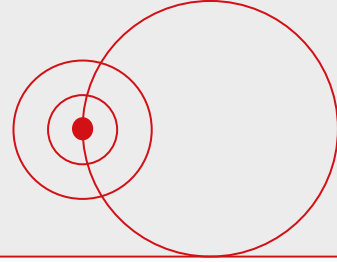


Phillip Kieval
(University of Florida)



Oscar Westerblad
(University of Iceland)

Different forms of understanding

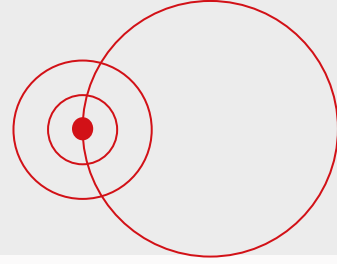


Pragmatic understanding

- A change of a scientist's opportunities for action - sometimes called 'affordance space'.
- **Affordance**: the perception of what the environment offers us, what it provides or furnishes, either for good or ill. (Gibson 2015)
- Pragmatic understanding of one's affordance space = having the skill of effectively using a device for goal-oriented action. (Stuart 2025)



Pragmatic understanding vs explanation



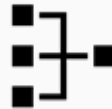
Explanatory understanding



Using concepts, theories, mathematical, material or DL models to explain how and why something happens.



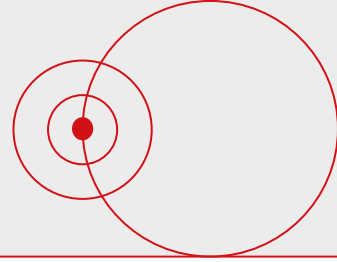
Pragmatic understanding



Understanding how ML can be best used to investigate phenomena.



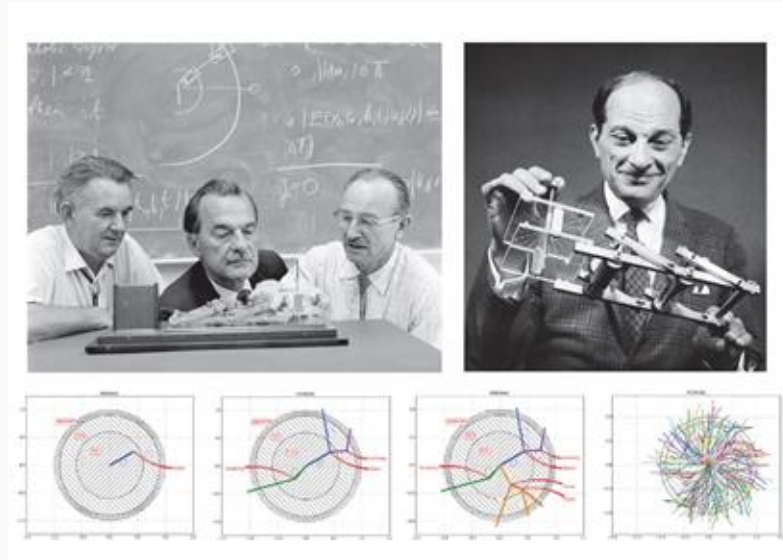
Different forms of Understanding

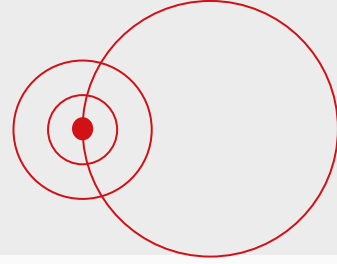


Scientists can properly use a new tool to generate meaningful research without a complete, technical understanding of it.

But rather with some **low-level understanding**:

- Trading zone (Peter Galison): simplified, shared practical language to coordinate actions with a device.





Periodic Table of Expertises

Ubiquitous Expertises

Dispositions

Interactive Ability

Reflective Ability

Specialist Expertises

Ubiquitous
Tacit Knowledge

Specialist
Tacit Knowledge

Beer Mat Knowledge Popular Understanding Primary Source Knowledge

Interactional Expertise Contributory Expertise

Polimorphic

Mimeomorphic

Meta-expertises

External
(Transmuted Expertises)

Internal
(Nontransmuted Expertises)

Ubiquitous Discrimination Local Discrimination

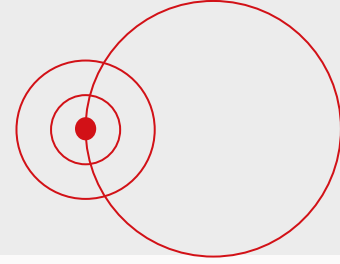
Technical Connoisseurship Downward Discrimination Referred Expertise

Metacriteria

Credentials Experience Track Record

Expertise (Collins and Evans 2007):

- **Interactional expertise:** the ability to interact meaningfully with experts in a domain even without the ability to contribute to the domain.
- **Internal meta-expertise:** the ability to identify the expertise of experts in a domain.



- Knowledge-how: strategies of using a tool, successes and failure cases, and evaluation standards; more important than accuracy, rigor and objective procedures.

Knowledge-how in models

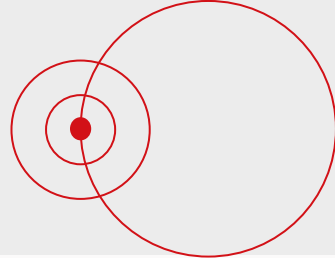
-  Instructions in canonical contexts.
-  Dos and don'ts.
-  Troubleshooting strategies.
-  Cases of new uses under different contexts.
-  A collection of conditions for uses or purposes.
-  Other elements used in combination.
-  Ways of evaluating performance.



Closing Figure An illustration of floating nests from *The Universe, Or, The Wonders of Creation: The Infinitely Great and the Infinitely Little* by [F.A. Pouchet](#) (1883). Illustration by A. Mesnel

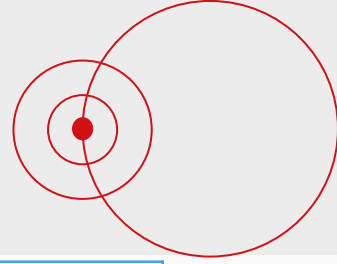
Source: Classic Image/Alamy Stock Photo.

Questions concerning pragmatic understanding



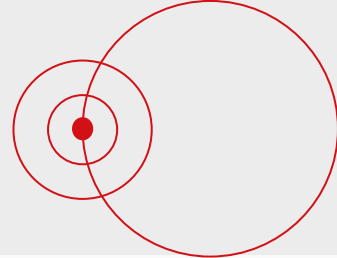
- **When can we say that scientists possess pragmatic understanding?**
 - A cluster of knowledge?
 - An ability to fulfill a purpose?
 - A skill that can be learned methodically?
- **What forms can pragmatic understanding of AI take?**
 - A remaining empirical question.
- **What makes pragmatic understanding plausible, reliable, generalizable?**
 - Repetitive practical success?
 - Being able to predict and explain the success?
 - A robust and diverse web of elements?
 - Forming systematic lessons across domains?

Interview Questions



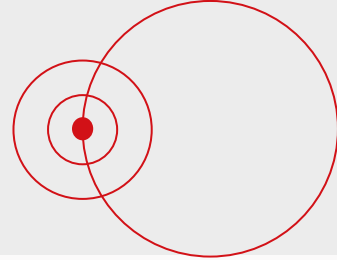
Topics	Sample questions
Research projects	What are your research projects? Does AI enhance your outcomes compared to traditional methods?
Academic background	What brought you to exploring AI in astronomy? What disciplines did you study and how did you learn to use AI?
AI experience	Describe experiences when you realized that AI is the right tool. Describe experiences when you had difficulties implementing AI or found its benefits limited.
Vision on progress	What are some most exciting research projects that use AI in your field? What are some projects that do not live up to the high expectations?
Interdisciplinary experience	How did you interact with computer scientists? What are some platforms that are most helpful for research? What career suggestions would you provide / have you received?

Interview Summaries



1. Tempered optimism

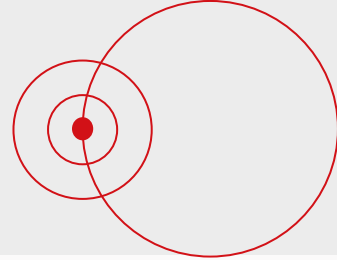
- AI can help us find patterns or make inferences from non-Gaussian fields, *but* we don't know how much new physics those places contain.
- AI can accelerate simulation, data processing and inference, *but* we need to consider whether this order of magnitude is worth pursuing.
- AI by itself cannot produce novel results. Scientists need to make sense out of AI's use and make the findings meaningful.



1. Tempered optimism

2. Awareness and avoidance of the hype

- One needs to know how many different tools are out there and choose the best ones for their purpose.
- Not all questions are suitable for ML. Traditional methods are more transparent, certain and cheap.
- The right use of ML is to use it for questions that can *only* be solved by it, but not by any other tools. One needs to consider how many of those questions there are in astronomy.
- One needs to justify why ML is necessary for their project.



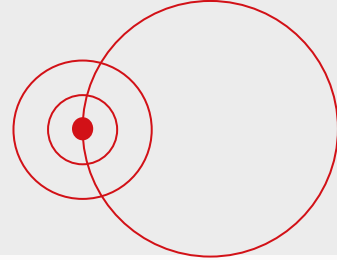
1. Tempered optimism

2. Awareness and avoidance of the hype

3. Pragmatic understandings of AI

a. AI as a statistical method

- ML is a natural extension of statistical methods.
- Beyond making inferences about patterns, correlations or parameters, it's important to quantify uncertainties, which can be hard for ML.
- Scientists should first study statistics well, especially familiarize themselves with statistical thinking.



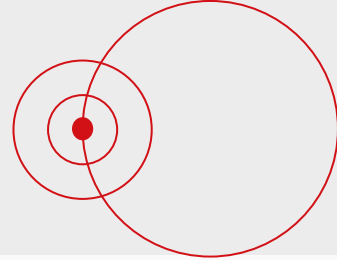
1. Tempered optimism

2. Awareness and avoidance of the hype

3. Pragmatic understandings of AI

- a. AI as a statistical method
- b. AI as an object of understanding and refinement

- **Physics of learning:** Physics may provide distinct perspectives on AI. The move from toy models to principles, and the use of statistical methods to study an object.
- **Experimentation on AI:** Scientists do not yet know how AI may perform as the training set increases. This is something to be found out by running experiments.
- **Special datasets:** AI is used on real datasets with special structures, size and designated aims.



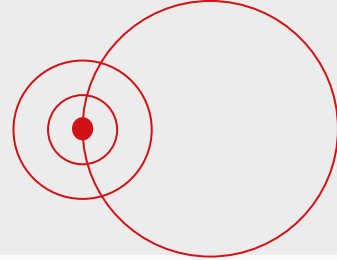
1. Tempered optimism

2. Awareness and avoidance of the hype

3. Pragmatic understandings of AI

- a. AI as a statistical method
- b. AI as an object of understanding and refinement
- c. Error repertoire

- There are certain mistakes that AI tends to make. One can learn about them and always keep an eye on them.



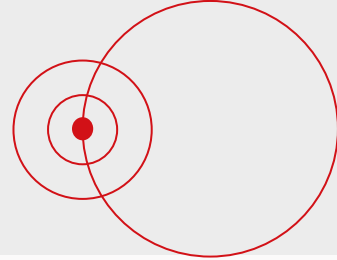
1. Tempered optimism

2. Awareness and avoidance of the hype

3. Pragmatic understandings of AI

- a. AI as a statistical method
- b. AI as an object of understanding and refinement
- c. Error repertoire
- d. Contextual design principles

- The choice of network structures should fit the structure of the data and the information that aim to be extracted.



1. Tempered optimism

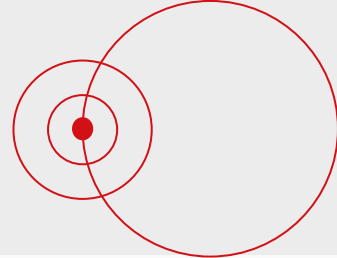
2. Awareness and avoidance of the hype

3. Pragmatic understandings of AI

- a. AI as a statistical method
- b. AI as an object of understanding and refinement
- c. Error repertoire
- d. Contextual design principles
- e. Interactional expertise

- Multiple astronomers reported benefitting from conversations with computer scientists.
- Some environments support projects from which computer scientists can also benefit.

Different styles of understanding



Theoretician: familiar with theories of ML, often with contributory expertise for both CS and specific fields.



Developer: absorbs and develops CS tools for the specific field. Might not collaborate much with CS.



Mediator: absorbs tools from both CS and the specific field. Often with interactional expertise and works in collaboration.

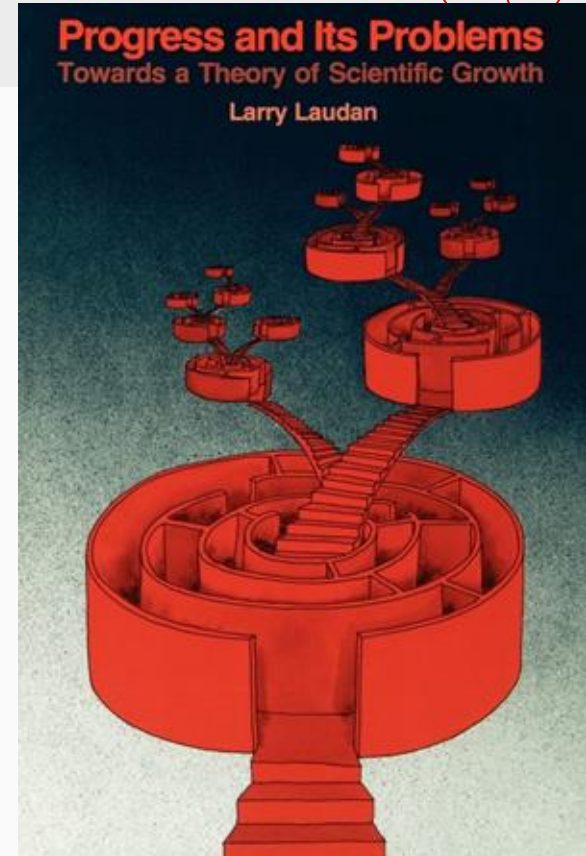
1. Tempered optimism
2. Awareness and avoidance of the hype
3. Pragmatic understandings of AI
4. Different working modes

Pursuit

I shall be arguing that a careful examination of scientific practice reveals that there are generally *two* quite different contexts within which theories and research traditions are evaluated.³⁸ I shall suggest that, within each of these contexts of

The context of acceptance. Beginning with the more familiar of the two, it is clear that scientists often choose *to accept* one among a group of competing theories and research traditions, i.e., *to treat it as if it were true*. Particularly in cases

The context of pursuit. Even if we had an adequate account of theory choice within the context of acceptance, however, we would still be very far from possessing a full account of rational appraisal. The reason for this is that there are many important situations where scientists evaluate competing theories by criteria which have nothing directly to do with the acceptability or “warranted assertibility” of the theories in question.



What makes an idea pursuitworthy?

Minimum plausibility

Interestingness

Fertility

Unificatory power

Explanatory Power

Simplicity/elegance

Peirce on practical costs

Proposals for hypotheses inundate us in an overwhelming flood ... the process of verification to which each one must be subjected before it can count as at all an item, even of likely knowledge, is so very costly in time, energy, and money—and consequently in ideas which might have been had for that time, energy, and money, that Economy would override every other consideration even if there were any other serious considerations. (Peirce 1932–58, 5.602)



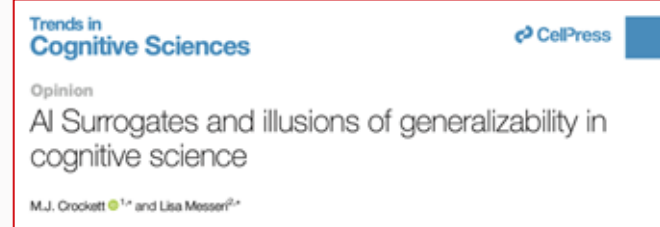
C. S. Peirce (1839 - 1914)

Pursuit risks

Spurious plausibility: regarding an idea as more plausible than it is

Illusory fertility: regarding an idea as more fertile/useful than it is

Efficiency traps: perceived speed of AI introduces perverse incentives about how to distribute scientific resources



The impact of AlphaFold2 one year on

The greatly improved prediction of protein 3D structure from sequence achieved by the second version of AlphaFold in 2020 has already had a huge impact on biological research, but challenges remain: the protein folding problem cannot be considered solved. We expect fierce competition to improve the method even further and new applications of machine learning to help illuminate proteomes and their many interactions.

David T. Jones and Janet M. Thornton

The Ideation–Execution Gap: Execution Outcomes of LLM-Generated versus Human Research Ideas

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Stanford University

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PNAS

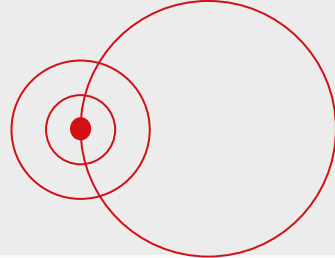
OPINION



Protein folds vs. protein folding: Differing questions, different challenges

Shi-Jie Chen¹, Mubashir Hassan², Robert L. Jernigan³, Kaijie Jia⁴, Daisuke Kihara⁵, Andrzej Kloczkowski⁶, Sergei Kotelnikov⁷, Dima Kozlov⁸, Jie Liang⁹, Adam Liver¹⁰, Silvina Matyska¹¹, Jerrel Meller¹², Cristian Micheletti¹³, Julie C. Mitchell¹⁴, Sayantan Mondal¹⁵, Ruth Nussinov¹⁶, Kei-ichi Okazaki¹⁷, Dmitry Padhorny¹⁸, Jeffrey Skolnick¹⁹, Yubin R. Sosnick²⁰, George Stan²¹, Ilya Vakser²², Xiaojin Zou²³, and George D. Rose²⁴

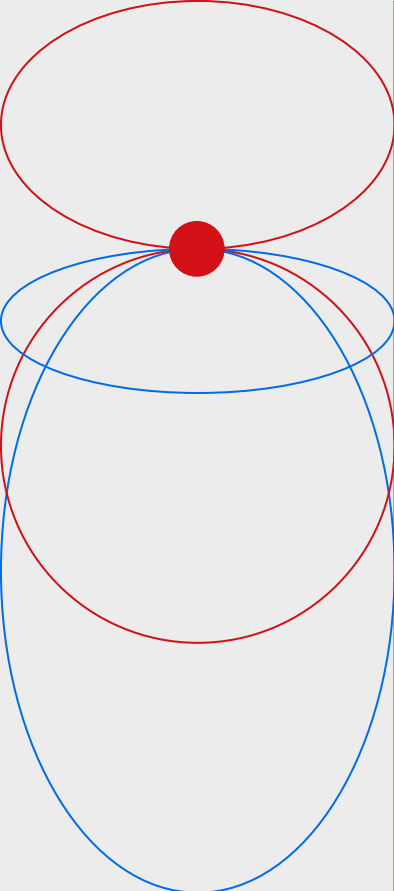
Conclusion & Discussion



Why does pragmatic understanding matter?

- Knowing how to use a tool reliably vs./+ detailed knowledge or explanation
- What contributes to the former - usefulness, know-how, perspectives, shared language and space, rules of thumb, expertise, experience ...?
- Bears on how to allocate valuable resources of time, energy, etc.

Your thoughts and experiences?



Thank you!

Want to keep the conversation going? We welcome new collaborators. Please get in touch!

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